

BREAKHOOK · PRECIS

BreakHook design precis

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
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Source	docs/02-concept.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-03	Amish Chadha	TRL 2 and TRL 3 on the constructable design; decisions under Amish's 2026-10-03 pre-approval (BHK-DDR-001, BHK-DDR-002); numbers from BHK-CAL-001
0.1	2026-09-30	Amish Chadha	Initial scaffold

BreakHook design precis

Lets residents pull a shack down from a safe distance to open a firebreak before a fire jumps across.

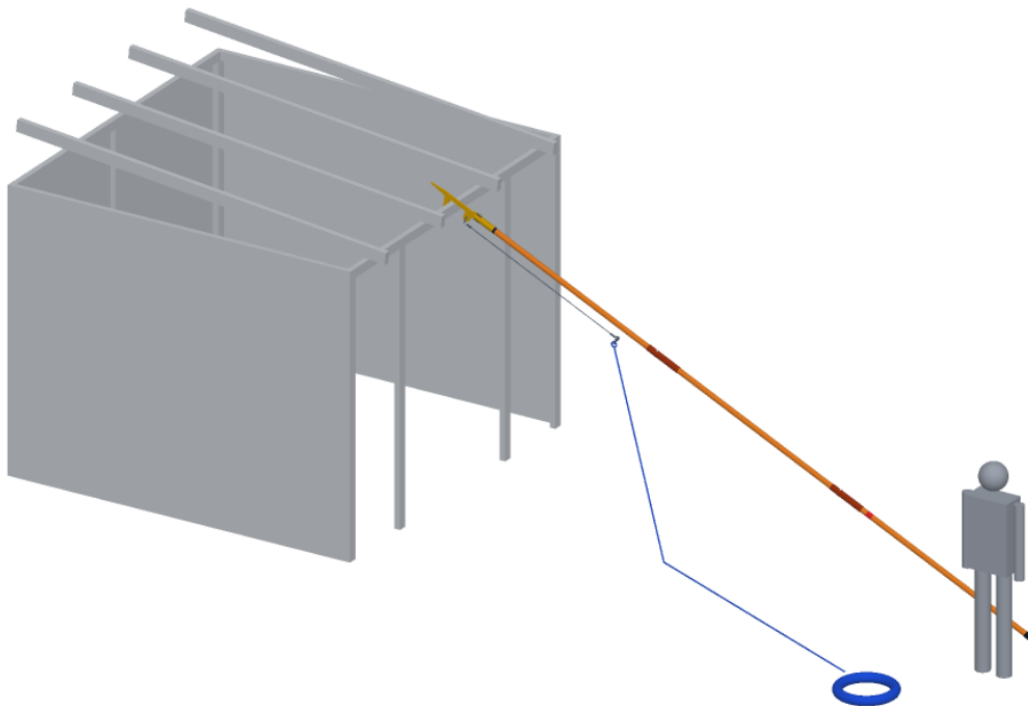
■ Summary

BreakHook is a steel hook on a 5.9 m fibreglass pole, with a wire rope leader and a 25 m pull rope. A team of two sets the hook over a wall plate or rafter from 4 m away, withdraws the pole and steps clear; a crew of about ten on hauling toggles, at least 10 m back, pulls the dwelling down away from the fire's path. A community kit holds two sets in a locked, sealed wall rack with check cards, fall zone tape and gloves. The kit's estimated cost is USD 1,098 against a USD 2,000 value-engineering target, and the pole and hook head weigh 7.82 kg.

BreakHook

github.com/BoujeeEnjinia1701/breakhook

CONCEPT, NOT FOR FABRICATION



Seen from the front right and above, 24 deg elevation. Hook set over the wall plate of a test frame, butt on the ground.
Grey: test frame and a 1.75 m person for scale

Figure 1. Concept render: the hook over the wall plate of a test dwelling frame, pole butt on the ground. Grey parts are context.

■ How it works

1. **Check.** The caller works through the check card: the dwelling is empty of people and animals (two people confirm), there is no overhead or informal wiring within 3 m of where the pole will go, the dwelling is free-standing or the end of a row, and it is at least two dwellings ahead of the fire.

2. **Mark.** Two people tape off a fall zone in front of the dwelling, at least as deep as the dwelling is tall, plus the hauling lane.
3. **Set.** The pole team joins the three sections (two nylon pins), pushes the hook head onto the tip, and lifts the hook over the wall plate or a rafter end. The front person lifts, the rear person holds the butt down, both hands below the red band.
4. **Withdraw.** A firm 50 N pull frees the pole from the hook head. The pole team carries the pole back out of the fall zone.
5. **Haul.** The pull rope runs from the hook head, through a short steel wire leader, back to the crew. Ten haulers on toggles, the nearest 10.8 m from the wall, take up the slack and pull on the caller's word. The frame racks or tips toward them and falls inside the taped zone.
6. **Recover.** The hook is pulled out of the debris by the rope when it is safe to approach.

BreakHook: pull force path at the working pull, kN (BHK-CAL-001 estimates)

github.com/BoujeeEnjinia1701/breakhook

CONCEPT, NOT FOR FABRICATION

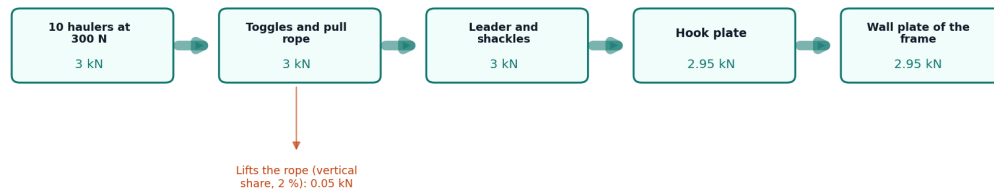


Figure 2. Force path at the 3 kN working pull (estimates from BHK-CAL-001).

■ Main components

Table 1. Components of one set (numbers match bom/bom.csv)

BOM	Component	What it is
1	Hook plate	10 mm S355 plate, profile cut: spike, 36 mm shank, arm raked back 17° that hangs 117 mm below the shank, rope tab with a 13 mm shackle hole
2	Socket tube	50.8 x 2.0 steel tube, 180 long, slotted across one end for the plate
3	Welding and finishing	Four 6 mm fillet welds; zinc-rich primer and yellow paint; WLL 3 kN and serial stamped
4	Pole sections (3)	Foam-filled, electrical-grade fibreglass tube 44.5 x 3.2, 1.8 m each, with the maker's ASTM F711 certificate
5, 6, 7	Joints (2)	Fibreglass sleeve 50.8 x 3.2 x 300 bonded to the lower section; nylon 10 mm pin through the upper section
8, 9, 10	Friction ring, hand band, butt cap	Rubber tape ring for a 50 N release; red band 1.55 m from the butt; rubber cap
11, 12	Shackles and leader	Two rated 3/8 in bow shackles; 1.5 m of 6 mm galvanised wire rope with thimble eyes
13	Pull rope	25 m of 12 mm polyester double braid, 25 kN or more, spliced eye
14, 15	Hauling toggles	Ten 300 mm fibreglass toggles on 6 mm prusik cords, 1 m apart from 11 m

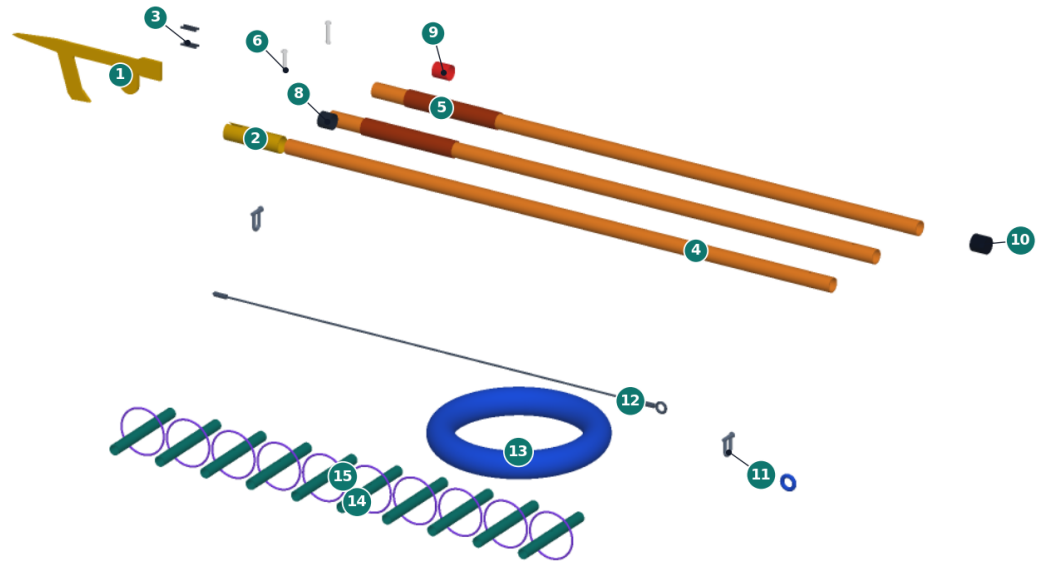
The community kit adds a wall rack (BOM 16, 17), paint (18), lock cable and numbered seals (19, 20), fall zone tape and stakes (21, 22), check cards (23), rigger gloves (24) and a proof load for each hook head (25).

BreakHook: exploded view

github.com/BoujeeEnjinia1701/breakhook

CONCEPT, NOT FOR FABRICATION

- 1 Hook plate
- 2 Socket tube
- 3 Socket welds
- 4 Pole sections (3)
- 5 Joint sleeves (2)
- 6 Joint pins (2)
- 8 Friction ring
- 9 Hand band
- 10 Butt cap
- 11 Shackles (2)
- 12 Wire rope leader
- 13 Pull rope
- 14 Hauling toggles
- 15 Prusik cords



Seen from the front right and above, 24 deg elevation; one set, pole sections laid side by side; numbers match bom/bom.csv

Figure 3. One set pulled apart; numbers match bom/bom.csv.

■ Key design choices

All made on 2026-10-03 under Amish's pre-approval; argued in BHK-DDR-001 (TRL 2 review) and BHK-DDR-002 (design for construction).

- **The pole places, the rope pulls.** The pole slides off the hook head and is withdrawn before the haul, so nobody holds anything inside the fall zone and the pole is never loaded by the pull.
- **Hook profile cut from one plate.** No forging; any welder with a plasma cutter, or a drill and grinder, can make it. The rope tab puts the pull in line with the arm, so the shank carries almost pure tension.
- **Foam-filled, certified fibreglass tube.** Electrical-grade tube is only insulating if its bore stays dry, hence foam fill, sealed ends, external sleeves and nylon pins. The pole is still never used within 3 m of overhead lines.
- **Wire rope leader.** The first 1.5 m from the hook is steel so heat and sheet edges at the dwelling do not cut the polyester rope.
- **No pulley in the base kit.** Ten toggles on prusik cords instead; a pulley needs an anchor that settlements rarely have.
- **Governance built into storage.** Locked rack, numbered seal, two keyholders and a use log answer the misuse concern in BHK-PRB-001.

■ Numbers from the TRL 3 calculations

Table 2. Key figures (BHK-CAL-001)

Quantity	Value
Pull to tip or rack the target dwelling	2.15 to 3.0 kN (estimate)
Working pull, proof load	3 kN, 6 kN
Haulers at 300 N	10
Nearest hauler from the wall	10.8 m
Least factor on yield in the hook plate at proof	2.3 (100 mm deep beam)
Rope, leader and shackle factors at the working pull	7.5, 6.0, 3.3
Pole and hook head mass	7.82 kg
Lift and hold-down forces for the pole team	252 N and 168 N
Hook droop at 25°	about 0.66 m
Insulated length	3.67 m
Deployment within 100 m of the store	3.0 min (estimate)
Value-engineering target	USD 2,000. Estimated cost of the constructable design: USD 1,098 (USD 902 under the target)

■ Patent design-arounds

From the preliminary patent, trademark and prior-art screen (not legal advice), all kept in the constructable design:

- Cite the historical fire-hook lineage; no live patent found close to the concept.
- Non-conductive tail on the pole: 3.67 m of foam-filled fibreglass with no metal below the hook head.
- Empty-shack check protocol before any pull: the first item on the check card.
- Name changed from GapHook to BreakHook to avoid GapHook Oy, a Finnish software company.
- Watch items from the preliminary screen, both answered in the design: overhead lines (3 m rule, insulated length) and the structure falling toward users (pole withdrawn, haulers beyond the fall zone).

■ Shared blocks

- CalRig proof-load for the hook head, leader and shackles (6 kN for 1 minute before issue).

■ Relationship to other lab projects

BreakHook is a Design Molecule Labs situational field hardware design. Its proof load uses the lab's CalRig rig where one is available; otherwise a rigging shop does it.

■ Safety

Safety: The hazards are a falling structure, a dwelling that is not empty, electrical wiring, a rope system failing under load, heat and smoke, and crowds. The safety stops are fixed by design and by the check card:

- Never pull until two people have confirmed the dwelling is empty of people and animals.
- Never raise the pole within 3 m of any overhead or informal line. The pole's insulated length is a second line of defence only; the hook, leader and shackles are metal and a wet rope conducts.
- The pole is withdrawn and the pole team is out of the fall zone before anyone hauls.
- Haulers stand at least 10 m from the wall and at least 1.5 times the dwelling's height away, on the side away from the fire, in gloves; nobody stands beside the rope line within the fall zone.
- Every hook head, leader and shackle set is proof-loaded to 6 kN before issue and after any seal break; a bent hook, kinked leader or frayed rope is taken out of service.
- Only free-standing dwellings or end units; never multi-storey or masonry buildings.
- Pulling a home down destroys it. The decision belongs to the community and its fire plan, not to the tool.

This design is published as an open engineering reference. It is not certified equipment.

■ Open questions

None open. Every TRL 1 question was decided under Amish's 2026-10-03 pre-approval; facts that can only be settled with real parts are listed in the design decisions register (BHK-DEC-001).